

LAB#1

INTRODUCTION TO SQL, DDL and DML commands.

OBJECTIVES:

- To get familiar with SQL
- To implement DDL commands (Create, Alter, Truncate and Drop).
- To implement DML commands (Select, Insert, Update and Delete).

LAB TASKS:

1. Create database and the table Employee using following schema.

(Emp_no, E_name, E_address, E_ph_no, Dept_no, Dept_name, Job_id, Salary).

Query:

```
*LAB01> Script-1 X
CREATE TABLE Employee (
  Emp_no INTEGER PRIMARY KEY,
  E_name TEXT,
  E_address TEXT,
  E_ph_no TEXT,
  Dept_no INTEGER,
  Dept_name TEXT,
  Job_id TEXT,
  Salary REAL
);
```

Output:

Name	Value
Finish time	Sun Mar 29 01:32:34 PKT 2026
Query	CREATE TABLE Employee (Emp_no INTEGER PRIMARY KEY, E_name TEXT, E_address TEXT, E_ph_no TEXT, Dept_no INTEGER, Dept_name TEXT, Job_id TEXT, Salary REAL)

2. Insert at least 10 Records in the above database with repetitive department names in the column Dept_name.

Query:

```
INSERT INTO Employee VALUES
(1, 'Ali', 'Lahore', '1111111111', 10, 'HR', 'HR01', 40000),
(2, 'Sara', 'Karachi', '2222222222', 20, 'IT', 'IT01', 60000),
(3, 'John', 'Islamabad', '3333333333', 10, 'HR', 'HR02', 45000),
(4, 'Ayesha', 'Multan', '4444444444', 30, 'Finance', 'FN01', 70000),
(5, 'Ahmed', 'Lahore', '5555555555', 20, 'IT', 'IT02', 55000),
(6, 'Zara', 'Karachi', '6666666666', 30, 'Finance', 'FN02', 65000),
(7, 'Usman', 'Faisalabad', '7777777777', 10, 'HR', 'HR03', 48000),
(8, 'Hina', 'Sialkot', '8888888888', 20, 'IT', 'IT03', 62000),
(9, 'Bilal', 'Peshawar', '9999999999', 30, 'Finance', 'FN03', 72000),
(10, 'Nida', 'Quetta', '1010101010', 10, 'HR', 'HR04', 50000);
```

Value
Sun Mar 29 01:39:50 PKT 2026
Sun Mar 29 01:39:50 PKT 2026
INSERT INTO Employee VALUES
(1, 'Ali', 'Lahore', '1111111111', 10, 'HR', 'HR01', 40000),
(2, 'Sara', 'Karachi', '2222222222', 20, 'IT', 'IT01', 60000),
(3, 'John', 'Islamabad', '3333333333', 10, 'HR', 'HR02', 45000),
(4, 'Ayesha', 'Multan', '4444444444', 30, 'Finance', 'FN01', 70000),
(5, 'Ahmed', 'Lahore', '5555555555', 20, 'IT', 'IT02', 55000),
(6, 'Zara', 'Karachi', '6666666666', 30, 'Finance', 'FN02', 65000),
(7, 'Usman', 'Faisalabad', '7777777777', 10, 'HR', 'HR03', 48000),
(8, 'Hina', 'Sialkot', '8888888888', 20, 'IT', 'IT03', 62000),
(9, 'Bilal', 'Peshawar', '9999999999', 30, 'Finance', 'FN03', 72000),
(10, 'Nida', 'Quetta', '1010101010', 10, 'HR', 'HR04', 50000)

3. Display all records with DNo 10.

```
SELECT * FROM Employee
WHERE Dept_no = 10;
```

no	E_name	E_address	E_ph_no	Dept_no	Dept_name	Job_id	Salary
1	Ali	Lahore	1111111111	10	HR	HR01	40000.0
3	John	Islamabad	3333333333	10	HR	HR02	45000.0
7	Usman	Faisalabad	7777777777	10	HR	HR03	48000.0
10	Nida	Quetta	1010101010	10	HR	HR04	50000.0

4. Display E_name, Dept_no. and Dept_name as Employee Name, Department Number and Department.

```
SELECT
    E_name AS Employee_Name,
    Dept_no AS Department_Number,
    Dept_name AS Department
FROM Employee;
```

Employee_Name	Department_Number	Department
Ali	10	HR
Sara	20	IT
John	10	HR
Ayesha	30	Finance
Ahmed	20	IT
Zara	30	Finance
Usman	10	HR
Hina	20	IT
Bilal	30	Finance
Nida	10	HR

5. Display all the names of departments in the organization but the names should not be repetitive.

```
SELECT DISTINCT Dept_name FROM Employee;
```

Dept_name
HR
IT
Finance

6. Add a new column 'Dep_Head' in the table and insert the names of HODs against each record.

```
ALTER TABLE Employee ADD COLUMN Dept_head TEXT;
```

Value
10
0.007s
Sun Mar 29 01:48:33 PKT 2026
Sun Mar 29 01:48:33 PKT 2026
ALTER TABLE Employee ADD COLUMN Dept_head TEXT

```
UPDATE Employee SET Dept_head = 'Ali Raza' WHERE Dept_name = 'HR';
UPDATE Employee SET Dept_head = 'Sara Khan' WHERE Dept_name = 'IT';
UPDATE Employee SET Dept_head = 'John Smith' WHERE Dept_name = 'Finance';
```

7. Suppose the HOD of HR department has been changed. Update the name of the HOD in every record by using single query.

```
UPDATE Employee
SET Dept_head = 'SAM ALTMAN'
WHERE Dept_name = 'HR';
```

8. Suppose the Employee number 5 has resigned from the company. Remove the record of that employee from table.

```
DELETE FROM Employee WHERE Emp_no = 5;
```

Statistics 1 SQL Terminal X

DELETE FROM Employee WHERE Emp_no = 5

9. Explore the datatype Date and implement in the above table with column name 'Hire_date'.

```
DELETE FROM Employee WHERE Emp_no = 5;
ALTER TABLE Employee ADD COLUMN Hire_date DATE;
SELECT * FROM Employee;
```

10. Explore the CONCAT() function and concat operator and implement it on the above database by joining any two relevant columns. Also state the limitation of this function if there is any.

```
SELECT E_name || ' - ' || Dept_name AS Employee_Info
FROM Employee;
```

Results 1 SQL Terminal X

FROM Employee

Employee_Info
Ali - HR
Sara - IT
John - HR
Ayesha - Finance
Zara - Finance
Usman - HR
Hina - IT
Bilal - Finance
Nida - HR

9 row(s) fetched.

11. Write queries and show output for the following

- Truncate the table
- Drop the table
- Drop the database

```
DELETE FROM Employee;
```

I am using SQLite

Here it Uses DELETE Instead of Truncate.(Its the same)

```
> DELETE FROM Employee
```

```
9 row(s) modified.
```

```
DROP TABLE Employee;  
SELECT * FROM Employee
```

Question No 12:

1.I Have Revised Basic Operations of SQL Like INSERT , WHERE on thing that hinders me About this query Sometimes I format of the Query.Revisited some more examples.

2.The most Challenging task would be remembering names and inserting Data Time Consuming.